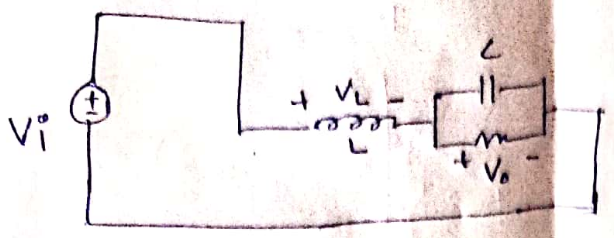
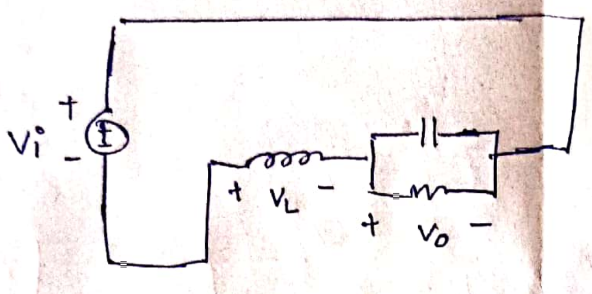
a) When switch is in position 1. for duration DT_s.



$V_i - V_L - V_o = 0$.

$V_L = \cancel{V_o - V_i}\ V_i - V_o \rightarrow (1)$.

When switch is in position 2. for duration $(1-D)T_s$.



$V_i + V_L + V_o = 0$.

$V_L = -V_o - V_i \rightarrow (2)$.

Applying voltsecond balance for a cycle.

$\cancel{(V_i - V_o)DT_s + (-V_o - V_i)(1-D)T_s = 0}$.

$(V_i - V_o)DT_s + (-V_o - V_i)(1-D)T_s = 0$.

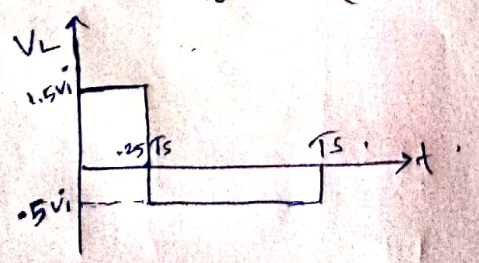
$\Rightarrow V_o = V_i(2D-1)$

b). voltage across inductor when $D = 0.25$

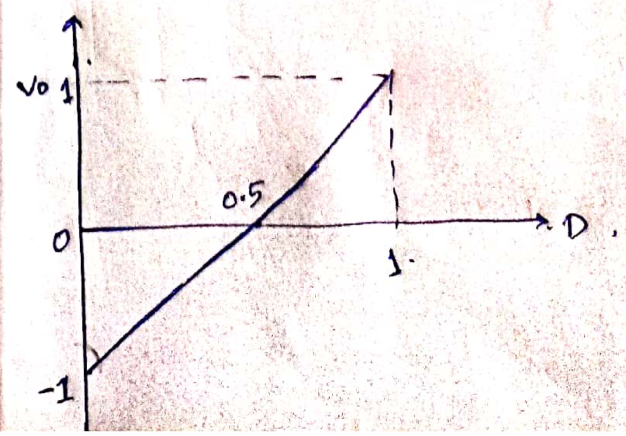
$V_o = V_i \times (2 \times .25 - 1) = -0.5\,V_i$

position (1) $\Rightarrow V_L = 1.5V_i$

position (2) $\Rightarrow V_L = -0.5V_i$.



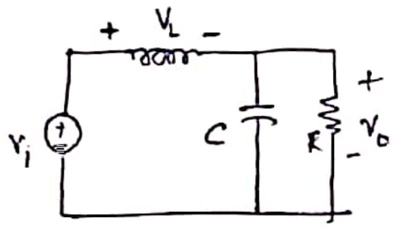
(c) $V_o = V_i(2D-1)$



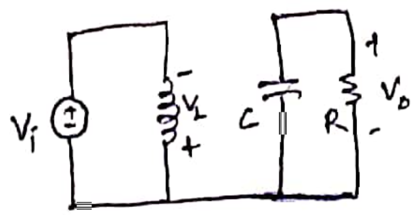
2.

(a)

During Position 1 circuit;



During Position 2 circuit:



Applying volt-sec Balance:

$$(V_i - V_o) D T_s + (-V_i)(1-D) T_s = 0$$

$$\frac{V_o}{V_i} = \frac{2D-1}{D}$$

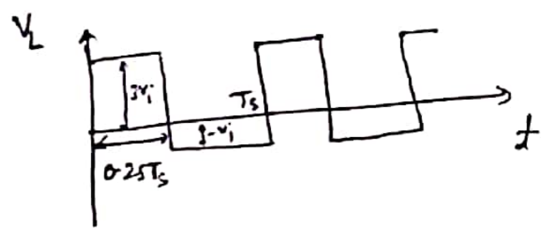
(b) voltage across Inductor, when $D = 0.25$

$$V_o = V_i \left(\frac{2D-1}{D} \right)$$

$$V_o = -2V_i$$

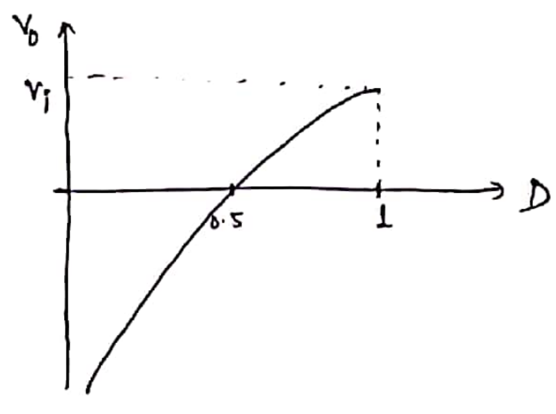
when switch in pos^n 1, $V_L = V_i - V_o = 3V_i$

" pos^n 2, $V_L = -V_i$



(c) $V_o = \left(\frac{2D-1}{D} \right) V_i$

for $0 \leq D \leq 1$



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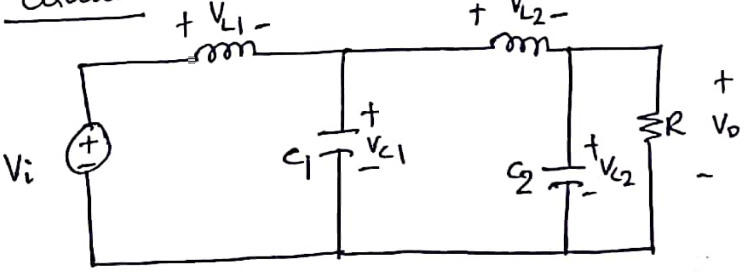
3.

(a). Switch in position (1) (DTs)

Circuit :-

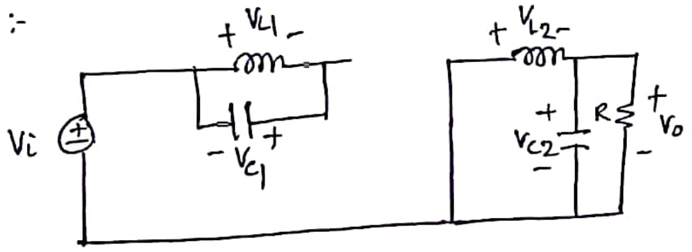


$V_{L1} = V_i - V_{C1}$ —— (1)

$V_{L2} = V_{C1} - V_o$ —— (2)

Switch in position (2) $(D'T_s)$ $[D' = 1 - D]$

Circuit :-



$V_{L1} = -V_{C1}$ —— (3)

$V_{L2} = -V_o$ —— (4)

Applying volt-sec balance in L_1,

eqn:: (1) and (3) $\Rightarrow$ $\langle V_{L1} \rangle = 0 \Rightarrow (V_i - V_{C1})D - V_{C1}D' = 0$

$V_i D - V_{C1}(D + D') = 0$

$\boxed{V_{C1} = V_i D}$

Volt-sec balance in L_2,

eqn:: (2) and (4), $(V_{C1} - V_o)D - V_o D' = 0$

$V_o = V_{C1} D$

$\boxed{V_o = D^2 V_i}$

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(b)

In DT_s

$V_{L1} = V_i - V_{c1} = V_i - DV_i = V_i(1-D)$

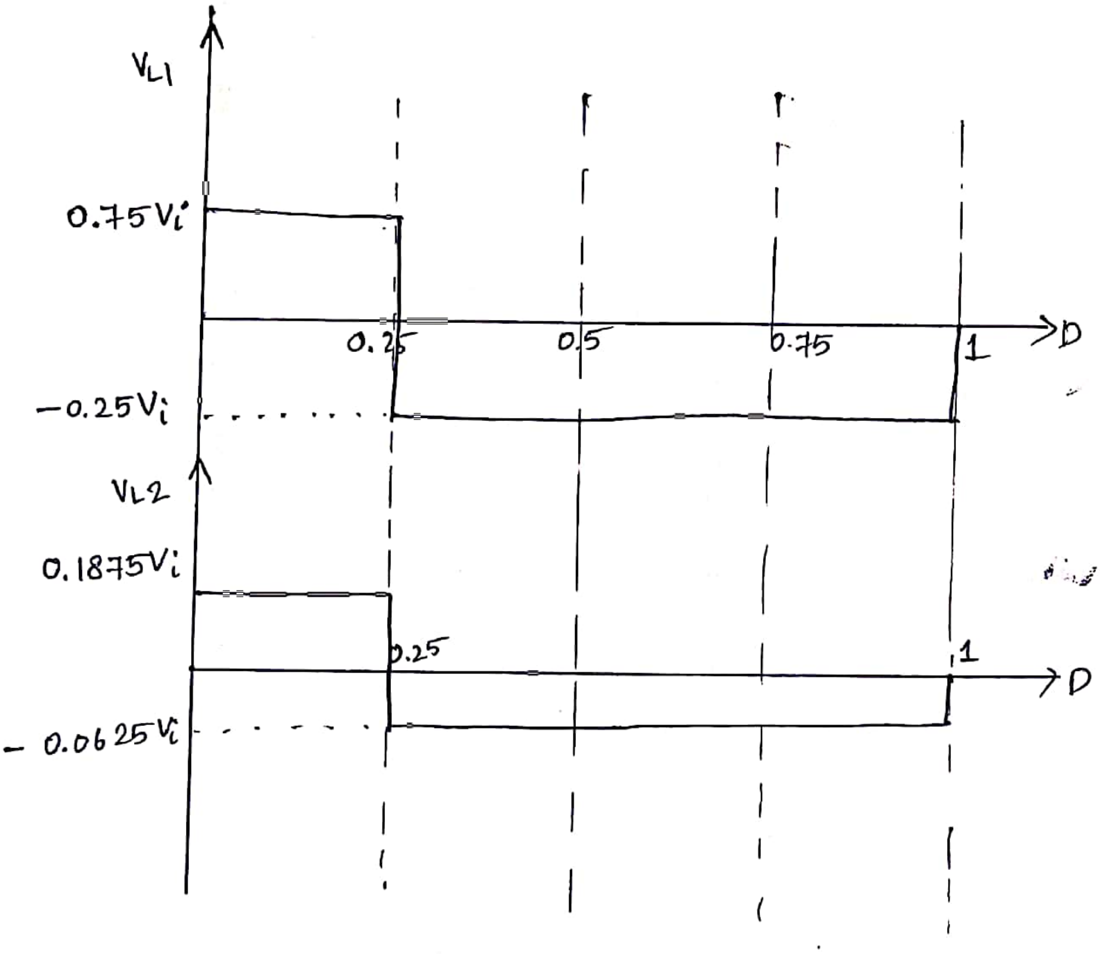
$V_{L2} = V_{c1} - V_o = DV_i - D^2V_i = V_iD(1-D)$

$D = 0.25$

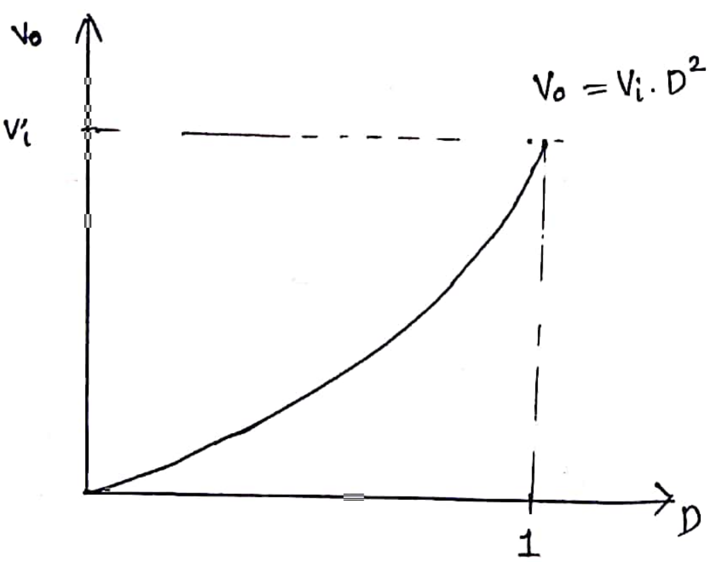
In $D'T_s$

$V_{L1} = -V_{c1} = -DV_i$

$V_{L2} = -V_o = -D^2V_i$



(c)



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